Welcome to NRES 746

Advanced Analysis Methods in Natural Resources and Environmental Science

Fall 2026

Instructor: Kevin Shoemaker
Office: FA 220E (see campus map)
Phone: (775)682-7449
Email: kevinshoemaker@unr.edu
Office hours: Mondays after class at 1pm in FA 220E (or by appointment)

Lecture: Mon, Wed at 9am in FA 109 (50 mins)
Lab: Tues at 3pm in FA 109 (2 hours 45 mins)

https://kevintshoemaker.github.io/NRES-746/

Course Description

Development and application of custom statistical models for the analysis of ecological and environmental data, including probability, likelihood, Bayesian and machine-learning approaches to inference, numerical estimation algorithms, hierarchical (multi-level) models, and models for non-independent observations such as generalized additive models, time series and spatial regression. Emphasis on translating ecological questions into statistical models and implementing custom solutions.

Course Objectives

Our focus will be on *model based inference*, including linear and non-linear regression, multi-level models, and spatial regression. We will use existing software tools where feasible, and we build our own **custom models** when needed. We use both classical (frequentist) and Bayesian methods. This course is designed to give you the tools and intuition needed to draw scientific conclusions from complex and messy observational data.

In this course, we are mathematicians. Many of us don't consider ourselves very 'mathy', and some of us shut down when we see mathematical notation. But it's impossible to overstate the value of mathematics for logical reasoning, problem solving and communicating methods unambiguously. There are no mathematics prerequisites for this course, but we will frequently use concepts from probability theory, calculus and linear algebra, since these are the main languages of statistics. Therefore, we will need to get comfortable with mathematical notation and symbol manipulation.

In this course, we are programmers. Knowing how to build custom statistical methods allows us to focus on our study systems and research questions, instead of being constrained by the software tools available to us. Mathematical equations give us the language of describing our custom models, but the solutions quickly become intractable to closed-form solutions. This is where the computer takes over- modern computational methods can get us pretty darn close to our target solutions- and that is usually more than enough. We just have to know a little programming to take advantage of these incredibly powerful methods (and collaborate effectively with our AI assistants!).

This is a safe space to learn and be dangerous Sometimes our research questions or intricacies of our study system lead us to dream up models that our 'canned' software can't handle easily. So it is very powerful to be able to build custom methods. When we do this, we are entering uncharted territory. And exploring can be dangerous… and your inner voices might tell you not to go there, just play it safe. Don't listen to those voices! In this class, you are allowed - and encouraged - to *be dangerous*! Just remember not to apply this motto uncritically outside the classroom!! Remember: science requires integrity and rigor above all. Therefore, we need to move slowly and cautiously in our role as scientific experts!

(NOTE: I encourage you to use existing software tools where possible because these methods have been rigorously tested over the years.)

Student Learning Outcomes

Upon completion of this course, students will be able to:

1. Identify and contrast the major classes of statistical methods used by natural resources scientists (e.g., Bayesian vs frequentist, likelihood-based, non-parametric methods, machine learning) and explain some of the key advantages and disadvantages of these methods.
2. Apply analysis tools such as logistic regression, non-linear regression, hierarchical (multi-level) models on diverse data sets representative of those commonly encountered in the natural resources sciences.
3. Learn to explore data sets quantitatively and graphically and to prepare data for analysis.
4. Perform command-line programming operations, statistical analysis, data visualization, and simulation modeling with the statistical computing language 'R'.
5. Critically evaluate the strength of inferences drawn from statistical models by testing assumptions and assessing performance.
6. Communicate concepts and methods via student-led lectures and discussion on advanced topics in data analysis.

Prerequisites

Curious scientific mind, broad research interests, and a willingness to engage with math and code. Students are expected to already have a basic foundation in mathematical and statistical concepts and methods, obtained through other coursework. If this is not the case, they should be prepared to work harder to develop the necessary prerequisite knowledge.

Textbooks and Readings

We will use the book, Ecological Models and Data in R, by Ben Bolker, as a general class reference. This is an older textbook but remains very relevant. It is also a good deskside reference that you are likely to return to in future work. There is an old PDF that Bolker makes available for free on his website: [https://math.mcmaster.ca/~bolker/emdbook/book.pdf](https://math.mcmaster.ca/~bolker/emdbook/book.pdf.

“The Ecological Detective” (by R. Hilborn and M. Mangel) is an older text written with similar goals and from a similar perspective to Bolker – we may have some readings and examples from this text as well.

We may occasionally use Simon Wood’s free ‘Core Statistics’ textbook as a more rigorous math-stats text: link here. I may also draw examples from Simon Wood’s “Generalized Additive Models: an introduction with R”.

Another recommended text (which I may draw some examples from) is Statistical Rethinking by Richard McElreath. This book takes an opinionated Bayesian perspective and keeps the math to a minimum, while covering some advanced statistical modeling approaches. This text also introduces Directed Acyclic Graphs (DAGs) and causal inference methods that are getting increased attention across many disciplines lately.

Additional readings may be assigned -please check the course schedule.

Other related books you may find useful include “A Primer of Ecological Statistics” by Gotelli and Ellison, “Bayesian Models: a Statistical Primer for Ecologists” by Hobbs and Hooten (recently updated), and the textbooks on Bayesian hierarchical models by Kery and Royle.

Course components

Class Participation: Students are expected to actively participate in the classroom. Don’t be afraid to ask questions- this is challenging (but fun!) material, and this classroom is a safe space for exploring and making mistakes! You are expected to come to class prepared to apply your knowledge (having completed any required reading). **Computers and AI tools are not allowed during class periods**- this is your chance to test your knowledge and ask lots of questions! You can enjoy plenty of time with your AI assistants outside of class!

Labs: Each week, we have a 165 min lab period, where you will be asked to complete a problem set that includes math problems, conceptual questions and coding-related questions. These problem sets are intended to be completed during the lab period, working both individually and in groups. Use of computers during lab sessions is not allowed- you are challenged to come to lab prepared and talk with your colleagues and instructor if you need help solving a problem. All lab submissions are made individually (even though you are encouraged to work closely with others).

Group Projects: Students will work on projects in groups of 2-3. Projects will require analysis of available data sets (e.g., unpublished data, reanalysis of previously published data, etc.) that are NOT intended to be part of a student’s thesis or dissertation chapters. The instructor can assist with identifying suitable data sets. Although a primary goal is to enhance knowledge and facility with the data analysis methods, an important secondary goal could be to develop a collaborative manuscript for publication! Therefore, careful thought should go into choice of a data set and relevant scientific questions to guide the analysis. The group project will take the form of a standard research manuscript. Unlike for labs, you are encouraged to use AI tools to help with things like literature review and data analysis (I still expect the writing to be your own, and you should fully acknowledge all AI use as part of the ms).

Group projects: expectations

Students will work in small groups to perform (and write up the results for) a data analysis using a custom, model-based analysis. Due to the nature of the class, you are asked to write up the model 'from scratch' rather than using a canned package, and pure 'machine learning' approaches are off limits unless explicitly approved by the instructor. The write-up will loosely take the format of a scientific paper to be submitted to a professional journal. However, because of the nature of this course, the most important pieces of the write-up are the methods and results sections. Nonetheless, I expect at least a few paragraphs introducing the topic and why it's important, and a few paragraphs discussing the implications of the results. The methods and results section can (and in many cases should) be longer than you typically see in a scientific paper- don't feel constrained by space for these sections! Not that you need to be wordy, I just want to make sure you have the space to clearly explain the analyses you performed and why you made the choices you did!

Here is a more detailed description of expectations for the final group project:

Abstract: Brief summary of project and main takeaways.

Introduction: Provide enough description so that the reader understands why the research is important and (if appropriate) what hypotheses or ideas are being tested.

Methods: Provide some details about data collection, just enough to give the reader the context necessary to understand the data. Provide plenty of detail about the analytical approach- enough detail so that a knowledgeable researcher could replicate the analysis. Justify all decisions that were made. Discuss key assumptions. Use equations!! This section can be longer than the methods section of a standard manuscript.

Results: Present all relevant results concisely. There is a limit of 5 figures and 3 tables, so choose carefully which figures and tables to present. Figures should be publication quality. Embed figures in the manuscript (usually methods and results sections) rather than putting all figures at the end.

Discussion: Write at least three paragraphs that put the results in a larger context and discusses areas of uncertainty. Potential topics are possible violations of assumptions, and future work that your analysis suggests would be profitable.

Code and data Provide all code (and ideally the raw data) used to run the analyses presented in the paper in a GitHub repository.

Acknowledgments: Use this section to acknowledge people who helped collect data, curate data, provide comments and guidance, etc.

AI statement: Use this section to document how generative AI tools were used in the project.

Literature cited: Exact format not important- but be consistent in whatever format you choose.

Grading

Course component	**Weight**
Laboratory exercises	40%
Participation and peer review	20%
Research Project	40%

Letter grade assignment is as follows: A (93 to 100%), A- (90 to 92.9%), B+ (87 to 89.9%), B (83 to 86.9%), B- (80 to 82.9%), C+ (77 to 79.9%), C (73 to 76.9%). Per Graduate School policy, grades below a C do not count toward a student's program of study at UNR.

Course Schedule

NOTE: the course schedule is subject to change, so please check back frequently!

Week	**Lecture 1**	**Lab**	**Lecture 2**	**Readings**
Aug. 24	Course introduction	Lab 1: Data visualizations, math/stats prerequisites	Algorithms in statistics	Bolker Ch 1-3, Bolker appendix 1
Aug. 31	Working with functions	Lab 2: Working with functions	Bestiary of functions	Bolker Ch. 3, Bolker appendix 2-4
Sept. 7	Probability	Lab 3: Working with probabilities	Bestiary of distributions	Bolker Ch. 4
Sept. 14	Synthetic data simulation	*Final project proposals- getting into groups*	Matrices and linear models	Bolker Ch. 5, Bolker appendix 8, Deep learning book ch 1
Sept. 21	Ordinary least squares	Lab 4: Data generating models	Intro to likelihood	Simon Wood textbook?
Sept. 28	Maximum likelihood estimation	Lab 5: Maximum likelihood estimation	Maximum likelihood estimation	Bolker Ch. 6,
Oct. 5	More likelihood	Lab 6: More likelihood!	Optimization	Bolker Chs. 6, 7
Oct. 12	Optimization	*Group meetings: final project proposals*	Markov Chain Monte Carlo (MCMC)	Bolker Ch. 7

Oct. 19	Bayesian inference	Lab 7: Bayesian inference	Model selection	Bolker Ch. 7
Oct. 26	Feature selection	Lab 8: Feature selection	Bayesian model selection	Bolker Ch. 6.8
Nov. 2	Modeling variance	Lab 9: Modeling variance	No class (veteran's day)	Bolker Ch. 10
Nov. 9	Bayesian multilevel models	Lab 10: Bayesian multilevel models	Bayesian multilevel models	Bolker Ch. 10
Nov. 16	Nonlinear regression and GAMs	*Group meetings: final project results*	Nonlinear regression and GAMs	Simon Wood textbook?
Nov. 23	Spatial regression models	Lab 11: Spatial regression models	Spatial regression models	Bolker Ch. 11
Nov. 30	Dynamic models	Lab 12: Dynamic models	Dynamic models	Bolker Ch. 11
Dec. 7	Class wrap-up- where to go from here	*Student project presenations!*	No class (prep day)	
Dec. 14	Project write-ups due by midnight Dec. 14			

Class Procedures

Students are expected to attend all lectures and labs in person. Where feasible, the instructor will attempt to record lectures and post to WebCampus for review but this class is not designed for remote or asynchronous learning. Attendance and engagement in class/labs is critical for learning. All assignments and labs should be submitted either directly to the instructor (for printed handouts) or using WebCampus. Deadlines for assignments can be found on WebCampus.

Communication

Outside of class, the best way to reach the instructor is by email (see Instructor Info above). Announcements, room changes, and schedule updates will be posted to WebCampus and/or the course website. Please check both regularly.

Illness Statement

If you are sick, please do not come to class. If you must miss class or lab due to illness, contact the instructor as soon as possible to make arrangements for any missed work or materials.

Make-up policy and late work:

If you miss a class meeting or lab period, it is your responsibility to talk to one of your classmates about what you missed. If you miss a lab meeting, you are still responsible for completing the lab activities and write-up by the stated due date. Please let your instructor know in advance if you need to miss class or lab.

University Policies

Statement on Academic Dishonesty

The University Academic Standards Policy defines academic dishonesty, and mandates specific sanctions for violations. See the University Academic Standards policy: UAM 6,502

Statement on Generative AI Use

Use of generative AI tools (e.g., ChatGPT, Claude, Copilot) is **not permitted** during class periods, labs, or on lab write-ups – these are meant to test your own understanding without outside assistance. For the group research project, AI tools are permitted for tasks such as literature review, debugging code, and improving writing clarity, but all AI use must be fully acknowledged in the manuscript's AI statement, and the scientific reasoning, analysis choices, and final writing must be your own. Outside of class, you are encouraged to use AI assistants to support your learning (e.g., working through practice problems, exploring code). Any unacknowledged use of AI in submitted work is a violation of the University Academic Standards Policy (UAM 6,502).

Statement of Disability Services

Any student with a disability needing academic adjustments or accommodations is requested to speak with the instructor or the Disability Resource Center (Pennington Achievement Center Suite 230) as soon as possible to arrange for appropriate accommodations. This course may leverage 3rd party web/multimedia content, if you experience any issues accessing this content, please notify your instructor.

Statement on Audio and Video Recording

Surreptitious or covert video-taping of class or unauthorized audio recording of class is prohibited by law and by Board of Regents policy. This class may be videotaped or audio recorded only with the written permission of the instructor. In order to accommodate students with disabilities, some students may have been given permission to record class lectures and discussions. Therefore, students should understand that their comments during class may be recorded.

Class sessions may be audio-visually recorded for students in the class to review and for enrolled students who are unable to attend live to view. Students who participate with their camera on or who use a profile image are consenting to have their video or image recorded. If you do not consent to have your profile or video image recorded, keep your camera off and do not use a profile image. Students who un-mute during class and participate orally are consenting to have their voices recorded. If you do not consent to have your voice recorded during class, keep your mute button activated and only communicate by using the “chat” feature, which allows you to type questions and comments live.

This is a safe space

The University of Nevada, Reno is committed to providing a safe learning and work environment for all. If you believe you have experienced discrimination, sexual harassment, sexual assault, domestic/dating violence, or stalking, whether on or off campus, or need information related to immigration concerns, please contact the University’s Equal Opportunity & Title IX office at 775-784-1547. Resources and interim measures are available to assist you. For more information, please visit the Equal Opportunity and Title IX page.

Statement on Campus Closures or Delays

In the event of class cancellations or delays caused by inclement weather conditions, fire/smoke conditions, or other unforeseen emergencies, the safety and well-being of students are the University’s top priority. Official notifications will be disseminated through the University website and other official channels with details related to any campus delays or closures. In the event of a campus closure, you will be informed as to whether the class will be offered remotely or if it will be canceled. If the class is cancelled, you will receive information on how to address any missed course content. Students facing significant impacts due to these events are encouraged to communicate with their instructor for potential accommodations.

Statement on Failure to Comply with Policy (including as outlined in this Syllabus) or Directives of a University Employee:

In accordance with section 6,502 of the University Administrative Manual, a student may receive academic and disciplinary sanctions for failure to comply with policy, including this syllabus, for failure to comply with the directions of a University Official, for disruptive behavior in the classroom, or any other prohibited action. “Disruptive behavior” is defined in part as behavior, including but not limited to failure to follow course, laboratory or safety rules, or endangering the health of others. A student may be dropped from class at any time for misconduct or disruptive behavior in the classroom upon recommendation of the instructor and with approval of the college dean. A student may also receive disciplinary sanctions through the Office of Student Conduct for misconduct or disruptive behavior, including endangering the health of others, in the classroom. The student shall not receive a refund for course fees or tuition.

Statement for Academic Success Services

Your student fees cover usage of the University Math Center, (775) 784-4433; University Tutoring Center, (775) 784-6801; and University Writing & Speaking Center, (775) 784-6030. These centers support your classroom learning; it is your responsibility to take advantage of their services. Keep in mind that seeking help outside of class is the sign of a responsible and successful student.

Mental Health Support Statement

There are times when you may experience difficulties in life, and you may benefit from seeking help. Mental health services are available to you as a student at no additional cost through Counseling Services at the Pennington Student Achievement Center. This includes same-day in-person and tele mental health initial consultations, brief individual counseling, and group counseling sessions. Limited same-day appointments can be scheduled online via Counseling Services or by calling 775-784-4648. Additional brief drop-in “Let’s Talk” student consultations are also available in the Counseling Services Annex located at the southwest corner of Great Basin Hall.

Veteran Statement

Veterans, Reservists, National Guard and military connected family members may wish to check the office of Veteran Services for benefits and support. Besides processing VA educational benefits, the department offers a variety of programs year-round to support student academic and personal success while transitioning to higher education and throughout your educational experience. They welcome inquiries regarding VA benefits and assist in navigating resources, the campus, and in the Reno community.